Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $C_r = {}^nC_r$, then value of $2\left(\frac{C_1}{C_0} + 2\frac{C_2}{C_1} + 3\frac{C_3}{C_2} + \cdots n\frac{C_n}{C_{n-1}}\right)$ is

(A) $n(n-1)$

(B) $n(n+1)$

(C) $n^2 - 1$

(D) $n^2 + 1$

Q.2

Suppose $[x]$ denote the greatest integer $\leq x$, and $n \in \mathbf{N}$, then $\lim_{n\to\infty} \frac{[{}^nC_o x^2] + [{}^nC_1 x^2] + \cdots + [{}^nC_n x^2]}{2^n - 2}$ is equal to

(A) $\frac{1}{2}x^2$

(B) x^2

(C) $2x^2$

(D) $4x^2$

Q.3

$\frac{C_0}{1} + \frac{C_2}{3} + \frac{C_4}{5} + \frac{C_6}{7} + \ldots\ldots\ldots =$

(A) $\frac{2^{n+1}}{n+1}$

(B) $\frac{2^{n+1}-1}{n+1}$

(C) $\frac{2^n}{n+1}$

(D) None of these

Q.4

The expression
$\left(\sqrt{2x^2+1} + \sqrt{2x^2-1}\right)^6 + \left(\frac{2}{\sqrt{2x^2+1}+\sqrt{2x^2-1}}\right)^6$ is a polynomial of degree:

(A) 5

(B) 6

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(C) 7

(D) 8

Q.5

The value of
$\left(\begin{array}{c} 30 \\ 0 \end{array} \right)\left(\begin{array}{c} 30 \\ 10 \end{array} \right) - \left(\begin{array}{c} 30 \\ 1 \end{array} \right)\left(\begin{array}{c} 30 \\ 11 \end{array} \right) + \left(\begin{array}{c} 30 \\ 2 \end{array} \right)\left(\begin{array}{c} 30 \\ 12 \end{array} \right) \ldots . + \left(\begin{array}{c} 30 \\ 20 \end{array} \right)\left(\begin{array}{c} 30 \\ 30 \end{array} \right)$
is where $\left(\begin{array}{c} n \\ r \end{array} \right) = {}^{n}C_{r}$

(A) $\left(\begin{array}{c} 30 \\ 10 \end{array} \right)$

(B) $\left(\begin{array}{c} 30 \\ 15 \end{array} \right)$

(C) $\left(\begin{array}{c} 60 \\ 30 \end{array} \right)$

(D) $\left(\begin{array}{c} 31 \\ 10 \end{array} \right)$

Q.6

Find the value of ${}^{4n}\mathrm{C}_0 + {}^{4n}\mathrm{C}_4 + {}^{4n}\mathrm{C}_8 + \ldots\ldots + {}^{4n}\mathrm{C}_{4n}$

(A) $(-1)^{\mathrm{n}}2^{2\mathrm{n}-1} + 2^{4\mathrm{n}-2}$

(B) $(-1)^{\mathrm{n}}2^{2\mathrm{n}-1}$

(C) $(-1)^{\mathrm{n}}2^{4\mathrm{n}-1}$

(D) None of these

Q.7

The sum of the series ${}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots\ldots - \ldots.. + {}^{20}C_{10}$ is

(A) 0

(B) ${}^{20}C_{10}$

(C) $-{}^{20}C_{10}$

(D) $\frac{1}{2}{}^{20}C_{10}$

Q.8

The value of ${}^{20}\mathrm{C}_0 + {}^{20}\mathrm{C}_1 + {}^{20}\mathrm{C}_2 + {}^{20}\mathrm{C}_3 + {}^{20}\mathrm{C}_4 + {}^{20}\mathrm{C}_{12} + {}^{20}\mathrm{C}_{13} + {}^{20}\mathrm{C}_{14} + {}^{20}\mathrm{C}_{15}$ is

(A) $2^{19} - \frac{\left({}^{20}\mathrm{C}_{10} + {}^{20}\mathrm{C}_9\right)}{2}$

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(B) $2^{19} - \frac{\left({}^{20}C_{10} + 2 \times {}^{20}C_9\right)}{2}$

(C) $2^{19} - \frac{{}^{20}C_{10}}{2}$

(D) none of these

Q.9

The sum $1 + \frac{1+a}{2!} + \frac{1+a+a^2}{3!} + \ldots \infty$ is equal to

(A) e^a

(B) $\frac{e^a - e}{a-1}$

(C) $(a-1)e^a$

(D) $(a+1)e^a$

Q.10

If $\sum_{r=0}^{n} \frac{r+2}{r+1} {}^nC_r = \frac{2^8 - 1}{6}$, then n is

(A) 8

(B) 4

(C) 6

(D) 5

Q.11

The value of $\sum_{r=0}^{n} {}^nC_r \sin(rx)$ is equal to

(A) $2^n \cdot \cos^n \frac{x}{2} \cdot \sin \frac{nx}{2}$

(B) $2^n \cdot \sin^n \frac{x}{2} \cdot \cos \frac{nx}{2}$

(C) $2^{n+1} \cdot \cos^n \frac{x}{2} \cdot \sin \frac{nx}{2}$

(D) $2^{n+1} \cdot \sin^n \frac{x}{2} \cdot \cos \frac{nx}{2}$

Q.12

For natural numbers m, n if $(1-y)^m (1+y)^n$
$= 1 + a_1 y + a_2 y^2 + \ldots$ and $a_1 = a_2 = 10$, then (m, n) is

(A) $(20, 45)$

(B) $(35, 20)$

(C) $(45, 35)$

(D) $(35, 45)$

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Q.13

The value of ${}^{10}C_0) + ({}^{10}C_0 + {}^{10}C_1) + ({}^{10}C_0 + {}^{10}C$
${}^{10}C_2) + \ldots + ({}^{10}C_0 + {}^{10}C_1{}^{10}C_2 + \ldots + {}^{10}C_9)$ is

(A) 2^{10}

(B) 10.2^9

(C) 10.2^{10}

(D) None of these

Q.14

The sum of the series
$\frac{2}{1} \cdot \frac{1}{3} + \frac{3}{2} \cdot \frac{1}{9} + \frac{4}{3} \cdot \frac{1}{27} + \frac{5}{4} \cdot \frac{1}{81} + \ldots\ldots\ldots\infty$ is equal to

(A) $\log_e 3 - \log_e 2$

(B) $\frac{1}{2} + \log_e 3 - \log_e 2$

(C) $\frac{1}{2} + \log_e 3 + \log_e 2$

(D) $\log_e 3 + \log_e 2$

Q.15

Which of the following is the greatest?

(A) ${}^{31}C_0{}^2 - {}^{31}C_1{}^2 + {}^{31}C_2{}^2 - \ldots - {}^{31}C_{31}{}^2$

(B) ${}^{32}C_0^2 - {}^{32}C_1^2 + {}^{32}C_1{}^2 - \ldots + {}^{32}C_{32}{}^2$

(C) ${}^{32}C_0^2 + {}^{32}C_1{}^2 + {}^{32}C_2{}^2 - \ldots + {}^{32}C_{32}2^2$

(D) ${}^{34}C_0{}^2 - {}^{34}C_1{}^2 + {}^{34}C_2{}^2 - \ldots + {}^{34}C_{32}{}^{32}$

Answers & Solutions

Q.1 Answer:

$n(n+1)$

Solution:

$\frac{C_r}{C_{r-1}} = \frac{n!}{r!(n-r)!} \cdot \frac{(r-1)!(n-r+1)!}{n!}$
$= \frac{n-r+1}{r}$
$\Rightarrow r\frac{C_r}{C_{r-1}} = n - r + 1$
$\Rightarrow 2\sum_{r=1}^{n} r\frac{C_r}{C_{r-1}} = 2\sum_{r=1}^{n}(n-r+1) = n(n+1)$

Q.2 Answer:

x^2

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Solution:

We know $x-1<[x]\leq x \forall x\in \mathbf{R}$, therefore ${}^{n}C_{k}x^{2}-1<\left[{}^{n}C_{k}x^{2}\right]\leq {}^{n}C_{k}x^{2}$

$\Rightarrow \quad \sum_{k=0}^{n}\left({}^{n}C_{k}x^{2}-1\right)<\sum_{k=0}^{n}\left[{}^{n}C_{k}x^{2}\right]\leq \sum_{k=0}^{n}{}^{n}C_{k}x^{2}$

$\Rightarrow \quad \frac{x^{2}(2^{n})-(n+1)}{2^{n}-2}<\frac{1}{2^{n}-2}\sum_{k=0}^{n}\left[{}^{n}C_{k}x^{2}\right]$

$\leq \frac{x^{2}(2^{n})}{2^{n}-2}$

Taking limit as $n\to\infty$, and using sandwich theorem, we get

$\lim_{n\to\infty}\frac{1}{2^{n}-2}\sum_{k=0}^{n}\left[{}^{n}C_{k}x^{2}\right]=x^{2}$

Q.3 Answer:

$\frac{2^{n}}{n+1}$

Solution:

Putting the value of $\mathrm{C}_{0},\mathrm{C}_{2},\mathrm{C}_{4}\ldots..$ we get

$=1+\frac{n(n-1)}{3.2!}+\frac{n(n-1)(n-2)(n-3)}{5.4!}+\ldots..=\frac{1}{n+1}$

$\left[(n+1)+\frac{(n+1)n(n-1)}{3!}+\frac{(n+1)n(n-1)(n-2)(n-3)}{5!}+\ldots.\right]$

Put $n+1=\mathrm{N}$

$=\frac{1}{N}\left[N+\frac{N(N-1)(N-2)}{3!}+\frac{N(N-1)(N-2)(N-3)(N-4)}{5!}+\ldots\right]$

$=\frac{1}{N}\left\{{}^{N}C_{1}+{}^{N}C_{3}+{}^{N}C_{5}+\ldots\right\}$

$=\frac{1}{N}\left\{2^{N-1}\right\}=\frac{2^{n}}{n+1}\quad\{\because N=n+1\}$

Q.4 Answer:

6

Solution:

$\frac{2}{\sqrt{2x^{2}+1}+\sqrt{2x^{2}-1}}=\sqrt{2x^{2}+1}-\sqrt{2x^{2}-1}$

$\therefore$ given expression

$=\left(\sqrt{2x^{2}+1}+\sqrt{2x^{2}-1}\right)^{6}+\left(\sqrt{2x^{2}+1}-\sqrt{2x^{2}-1}\right)^{6}$

we know that,

$(a+b)^{6}+(a-b)^{6}=2\left[a^{6}+{}^{6}\mathrm{C}_{2}a^{4}b^{2}+{}^{6}\mathrm{C}_{4}a^{2}b^{4}+{}^{6}\mathrm{C}_{6}b^{6}\right.$

$\therefore \left(\sqrt{2x^{2}+1}+\sqrt{2x^{2}-1}\right)^{6}+\left(\sqrt{2x^{2}+1}-\sqrt{2x^{2}-1}\right)^{6}$

$=2\left[\left(2x^{2}+1\right)^{3}+15\left(2x^{2}+1\right)^{2}\left(2x^{2}-1\right)\right.$

$\left.+15\left(2x^{2}+1\right)\left(2x^{2}-1\right)+\left(2x^{2}-1\right)^{3}\right]$

Which is a polynomial of degree 6.

Q.5 Answer:

$\begin{pmatrix}30\\10\end{pmatrix}$

Solution:

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To find

$^{30}C_0{}^{30}C_{10} - {}^{30}C_1{}^{30}C_{11} + {}^{30}C_2{}^{30}C_{12} - \ldots {}_{11}{}^{30}C_{20}{}^{30}C_{30}$

We know that

$(1+x)^{30} = {}^{30}C_0 + {}^{30}C_1 x + {}^{30}C_2 x^2$

$+ \ldots + {}^{30}C_{20}x^{20} + \ldots {}^{30}C_{30}x^{30}$ **...(1)**

$(x-1)^{30} = {}^{20}C_0 x^{30} - {}^{30}C_1 x^{29} + \ldots . + {}^{30}C_{10}x^{20}$

$-{}^{30}C_{11}x^{19} + {}^{30}C_{12}x^{18} + \ldots {}^{30}C_{30}x^0$ **...(2)**

Multiplying eq ${}^{n}(1)$ **and** (2) **and equating the coefficients of** x^{20} **on both sides, we get**

$^{30}C_{10} = {}^{30}C_0{}^{30}C_{10} - {}^{30}C_1{}^{30}C_{11} + {}^{30}C_2{}^{30}C_{12} - \ldots$

$+{}^{30}C_{20}{}^{30}C_{30}$

$\therefore$ **Req. value is** $^{30}C_{10}$

Q.6 Answer:

$(-1)^{n}2^{2n-1} + 2^{4n-2}$

Solution:

$(1+x)^{4n} = 4n^n C_0 + {}^{4n}C_1 x + {}^{4n}C_2 x^2 + {}^{4n}C_3 x^3 + {}^{4n}C_4 x^4 + \ldots .. + {}^{4n}C_{un}x^{4n}$

Put $x = 1$ **and** $x = -1$**, then adding.**

$2^{4n-1} = {}^{4n}C_0 + {}^{4n}C_2 + {}^{4n}C_4 + \ldots .. + {}^{4n}C_{4n}$ **..... (i)**

Now put, $x = i$

$(1+i)^{4n} = {}^{4n}C_0 + {}^{4n}C_1 i - {}^{4n}C_2 - {}^{4n}C_3 i + {}^{4n}C_4 + \ldots \ldots .. + {}^{4n}C_{4n}$

Compare real and imaginary part, we get

$(-1)^n (2)^{2n} = {}^{4n}C_0 - {}^{4n}C_2 + {}^{4n}C_4 - {}^{4n}C_6 + \ldots . + {}^{4n}C_{4n} \ldots$ **(ii)**

Adding (i) and (ii), we get $\Rightarrow \quad {}^{4n}C_0 + {}^{4n}C_4 + \ldots . + {}^{4n}C_{4n} = (-1)^n (2)^{2n-1} + 2^{4n-2}$

Q.7 Answer:

$\frac{1}{2}{}^{20}C_{10}$

Solution:

We know that, $(1+x)^{20} = {}^{20}C_0 + {}^{20}C_1 x + {}^{20}C_2 x^2$

$+ \ldots ..{}^{20}C_{10}x^{10} + \ldots .{}^{20}C_{20}x^{20}$

Put $x = -1, (0) = {}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots ...$

$^1 + {}^{20}C_{10}^2 - {}^{20}C_1^3 \ldots . + {}^{20}C_{20}$

$\Rightarrow 0 = 2\left[{}^{20}C_0 - {}^{20}C_1 + {}_{20}{}^{20}C_2 - {}^{20}C_3 + \ldots .. - {}^{20}C_9\right] + {}^{20}C_{10}$

$\Rightarrow {}^{20}C_0 - {}^{20}C_1 + {}^{20}C_2 - {}^{20}C_3 + \ldots . + {}^{20}C_{10} = \frac{1}{2}{}^{20}C_{10}$

Q.8 Answer:

$2^{19} - \frac{\left({}^{20}C_{10} + 2 \times {}^{20}C_9\right)}{2}$

Solution:

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$^{20}C_0 + {}^{20}C_1 + {}^{20}C_2 + {}^{20}C_4 + {}^{20}C_{12} + {}^{20}C_{13} + {}^{20}C_{14} + {}_{x} = {}^{20}C_{15}$

C_{n-x}

$^{20}C_{15} = {}^{20}C_5$

$C_{14} = {}^{20}C_6$

$^{20}C_{13} = {}^{20}C_7$

$^{20}C_{12} = {}^{20}C_8$

$\therefore$ we get$^{20}C_0 + C_1 + C_2 + C_4 + C_5 + C_6 + C_7 + C_8$

$\because {}^{20}C_0 + C_1 + \ldots C_8 + C_9 + C_{11} + \ldots {}^{20}C_{20} = 2^{20}$

$2\{C_0 + C_1 + \ldots C_9\} + C_{10} = 2^{20}$

$2\{C_0 + C_1 + \ldots C_8\} + {}^2C_9 + C_{10} = 2^{20}$

$\therefore C_0 + C_1 \ldots . C_8 = \frac{2^{20} - \left({}^2C_9 + C_{10}\right)}{2}$

$= 2^{19} - \frac{{}^{20}C_{10} + 2^{20}C_9)}{2}$

Q.9 Answer:

$\frac{e^a - e}{a-1}$

Solution:

The given series is

$1 + \frac{1+a}{2!} + \frac{1+a+a^2}{3!} + \frac{1+a+a^2+a^3}{4!} + \ldots$

Here $T_n = \frac{1+a+a^2+a^3+\ldots. \text{ to n terms}}{n!}$

$= \frac{1(1-a^n)}{(1-a)(n!)} = \frac{1}{1-a}\left(\frac{1-a^n}{n!}\right)$

$\therefore T_1 + T_2 + T_3 + \ldots .$ **to** ∞

$= \frac{1}{1-a}\left[\frac{1-a}{1!} + \frac{1-a^2}{2!} + \frac{1-a^3}{3!} + \ldots . \textbf{ to } \infty\right]$

$= \frac{1}{1-a}\left[\left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \ldots . to\infty\right) - \left(\frac{a}{1!} + \frac{a^2}{2!} + \frac{a^3}{3!} + \ldots . \textbf{ to } \infty\right)\right]$

$= \frac{1}{1-a}\left[(e-1) - (e^a - 1)\right] = \frac{e - e^a}{1-a} = \frac{e^a - e}{a-1}$

Q.10 Answer:

5

Solution:

$\sum_{r=0}^{n} \frac{r+2}{r+1} {}^nC_r = \sum_{r=0}^{n} \frac{r+1+1}{r+1} {}^nC_r$

$= \sum_{r=0}^{n} {}^nC_r + \sum_{r=0}^{n} \frac{{}^nC_r}{r+1} = 2^n + \sum_{r=0}^{n} \frac{{}^{n+1}C_{r+1}}{n+1}$

$= 2^n + \frac{1}{n+1}\sum_{r=0}^{n} {}^{n+1}C_{r+1}$

$= 2^n + \frac{1}{n+1}\left(2^{n+1} - 1\right)$

$= \frac{1}{n+1}\left[(n+1)2^n + 2^{n+1} - 1\right]$

$= \frac{1}{n+1}\left[2^n(n+3) - 1\right]$

Given, $\frac{(n+3)2^n - 1}{n+1} = \frac{2^8 - 1}{6} = \frac{(5+3)\cdot 2^5 - 1}{5+1}$

$\Rightarrow n = 5$

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Q.11 Answer:

$2^n \cdot \cos^n \frac{x}{2} \cdot \sin \frac{nx}{2}$

Solution:

$$\sum_{r=0}^{n} {}^nC_r \sin rx = Im\left(\sum_{r=0}^{n} {}^nC_r e^{irx}\right)$$

$$= Im\left(\sum_{r=0}^{n} {}^nC_r \left(e^{ix}\right)^r\right) = \operatorname{Im}\left(\left(1+e^{ix}\right)^n\right)$$

$$= Im(1+\cos x + i\sin x)^n$$

$$= \operatorname{Im}\left(2\cos^2\frac{x}{2} + 2i\sin\frac{x}{2}\cdot\cos\frac{x}{2}\right)^n$$

$$= \operatorname{Im}\left(2\cos\frac{x}{2}\left(\cos\frac{x}{2} + i\sin\frac{x}{2}\right)\right)^n$$

$$= 2^n \cdot \cos^n\frac{x}{2} \cdot \sin\frac{nx}{2}$$

Q.12 Answer:

$(35, 45)$

Solution:

$$(1-y)^m(1+y)^n$$

$$= \left[1 - {}^m\mathrm{C}_1 y + {}^m\mathrm{C}_2 y^2 - \ldots\ldots\right]\left[1 + {}^n\mathrm{C}_1 y + {}^n\mathrm{C}_2 y^2 + \ldots.\right]$$

$$= 1 + (n-m)y + \left\{\frac{m(m-1)}{2} + \frac{n(n-1)}{2} - mn\right\} y^2 + \ldots.$$

$$= 1 + (n-m)y + \left\{\frac{m(m-1)}{2} + \frac{n(n-1)}{2} - mn\right\} y^2 + \ldots.$$

By comparing coefficients with the given expression, we get

$\therefore a_1 = n - m = 10$ **and**

$a_2 = \frac{m^2+n^2-m-n-2mn}{2} = 10$

So, $n - m = 10$ **and** $(m-n)^2 - (m+n) = 20$

$\Rightarrow m + n = 80$

$\therefore m = 35, n = 45$

Q.13 Answer:

10.2^9

Solution:

$$\left({}^{10}C_0\right) + \left({}^{10}C_0 + {}^{10}C_1\right) + \left({}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2\right)$$

$$+ \ldots + \left({}^{10}C_0 + {}^{10}C_1 + {}^{10}C_2 + \ldots + {}^{10}C_9\right)$$

$$= 10\,{}^{10}C_0 + 9\,{}^{10}C_1 + 8\,{}^{10}C_2 + \ldots + {}^{10}C_9$$

$$= {}^{10}C_1 + 2\,{}^{10}C_2 + 3\,{}^{10}C_3 + \ldots + 10\,{}^{10}C_{10}$$

$$= \sum_{r=1}^{10} r\,{}^{10}C_r = 10\sum_{r=1}^{10} {}^9C_{r-1} = 10 \cdot 2^9$$

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Q.14 Answer:

$\frac{1}{2}+\log_e 3-\log_e 2$

Solution:

$$\frac{2}{1}\cdot\frac{1}{3}+\frac{3}{2}\cdot\frac{1}{9}+\frac{4}{3}\cdot\frac{1}{27}+\frac{5}{4}\cdot\frac{1}{81}+\ldots.\infty$$

$$=(1+1)\frac{1}{3}+\left(1+\frac{1}{2}\right)\left(\frac{1}{3}\right)^2+\left(1+\frac{1}{3}\right)\left(\frac{1}{3}\right)^3+\left(1+\frac{1}{4}\right)\left(\frac{1}{3}\right)^4+\ldots\infty$$

$$=\left\{\tfrac{1}{3}+\left(\tfrac{1}{3}\right)^2+\left(\tfrac{1}{3}\right)^3+\ldots\infty\right\}+\left\{\left(\tfrac{1}{3}\right)+\tfrac{1}{2}\left(\tfrac{1}{3}\right)^2+\tfrac{1}{3}\left(\tfrac{1}{3}\right)^3+\ldots.\infty\right\}$$

$$=\frac{\frac{1}{3}}{1-\frac{1}{3}}-\log_e\left(1-\tfrac{1}{3}\right)=\tfrac{1}{2}-\log_e\left(\tfrac{2}{3}\right)$$

$$=\tfrac{1}{2}+\log_e 3-\log_e 2$$

Q.15 Answer:

${}^{32}C_0^2+{}^{32}C_1{}^2+{}^{32}C_2{}^2-\ldots+{}^{32}C_{32}2^2$

Solution:

We know that ${}^nC_0{}^2+{}^nC_1{}^2+\ldots+{}^nC_n{}^2=2^nC_n$

and ${}^nC_0{}^2-{}^nC_1{}^2+\ldots+{}^nC_n{}^2$

$$=\begin{cases}0, & \text{if } n \text{ is odd}\\ {}^nC_{n/2}(-1)^{n/2}, & \text{if } n \text{ is even}\end{cases}$$

From this ${}^{31}C_0^2-{}^{31}C_0{}^2+{}^{31}C_2{}^2-\ldots-{}^{31}C_{31}{}^2=0$

${}^{32}C_0{}^2-{}^{32}C_1{}^2+{}^{32}C_2{}^2-\ldots+{}^{32}C_{32}{}^2=-{}^{32}C_{16}$

${}^4C_0{}^2-{}^{34}C_1{}^2+{}^{34}C_2{}^2-\ldots+{}^{34}C_{32}{}^2=-{}^{34}C_{17}$

${}^2C_0{}^2+{}^{32}C_1{}^2+{}^{32}C_2{}^2-\ldots+{}^{32}C_{32}{}^2={}^{64}C_{32}$

Obviously ${}^{64}C_{32}$ **is greatest.**

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